

What Makes a Headline Click? Linguistic Predictors of Click-Through Rate in Upworthy’s Headline A/B Tests

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1 Introduction

Between 2013 and 2015, Upworthy pioneered large-scale headline A/B testing for digital media. Each article was assigned two to six headline variants displayed simultaneously to random audience subsets, creating a natural experiment in which content is held fixed while framing varies. The resulting Upworthy Research Archive [1] documents approximately 22,000 such experiments, offering rare identification of headline effects on reader engagement.

This paper asks: *what linguistic and structural characteristics of a headline predict its click-through rate (CTR)?* Answering this question has direct implications for content editors, who make headline decisions under time pressure with limited systematic evidence.

We fit a cluster-robust OLS regression on 54,956 headline packages and find that emotional framing consistently suppresses CTR: exclamation marks reduce it by approximately 25%, collective “we/our” framing by 22%, and question headlines by 15%. Structural numeric cues produce modest but robust positive effects. Positive sentiment, measured by VADER compound scores [2], is associated with *lower* CTR, consistent with a negativity bias in online news consumption. These findings are stable across heteroskedastic-robust and impression-weighted specifications and do not vary significantly by content category.

2 Data

Source

We use the confirmatory split of the Upworthy Research Archive [1]. The unit of observation is a *package*: a unique headline variant paired with an article, characterized by the number of impressions served and clicks received. Packages nest within *tests* (each test is one article with multiple competing headlines shown to disjoint random audience subsets) and span the years 2013 to 2015. After removing single-variant tests, image-confounded tests, and a window affected by a randomization failure (details in Appendix A), the analytic sample contains 54,956 packages across 11,649 tests.

Outcome variable

Click-through rate is defined as $CTR_i = \text{clicks}_i / \text{impressions}_i$. Raw CTR is right-skewed and bounded in $[0, 1]$; we model $y_i = \log(CTR_i)$ throughout. The log transformation approximately normalizes the distribution and renders coefficients interpretable as proportional changes in CTR: for a binary feature with coefficient $\hat{\beta}_k$, the implied percentage change in CTR is $(e^{\hat{\beta}_k} - 1) \times 100\%$.

Predictors

We engineer 14 headline-level features from the raw text: eight binary structural and rhetorical indicators (ends with “?”, has “!”, has “...”, starts with a digit, contains any digit, starts with *how/why/what*, contains a superlative word, contains a negative word); two binary framing indicators (contains *you/your*, *we/our*); and three continuous readability measures (character count, word count, average word length). Sentiment is captured by the VADER compound score [2], a lexicon-based measure in $[-1, 1]$ requiring no model training. Two controls complete the specification: a year trend centered at 2013 to absorb platform-level changes in audience behavior over time, and the number of headline variants per test to account for within-test competition effects.

3 Final Regression Model

Specification

Let $y_i = \log(\text{CTR}_i)$ denote the outcome for package i and let $\mathbf{x}_i \in \mathbb{R}^{16}$ collect the 14 headline features and two controls described in Section 2. The model is:

$$y_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad i = 1, \dots, 54,956, \quad (1)$$

estimated by ordinary least squares.

Standard errors

Packages within the same A/B test share an audience pool, a publication window, and an underlying article, so their errors are likely correlated within tests. Ignoring this structure understates uncertainty. We report cluster-robust standard errors [3], which allow arbitrary error correlation among packages belonging to the same test while maintaining asymptotic validity across the $G = 11,649$ clusters.

To assess whether heteroskedasticity is an additional concern, we also compute HC3 heteroskedasticity-robust standard errors [4]. Across all 16 predictors the HC3-to-OLS standard error ratio ranges from 0.95 to 1.03, indicating that heteroskedasticity is negligible. Clustering, by contrast, inflates standard errors by factors of 1.3 to 2.2 relative to OLS, with the largest inflation for features that do not vary within a test (year and number of variants), confirming that within-test correlation is the dominant source of non-independence. Cluster-robust standard errors are therefore the basis for all inference reported below.

4 Discussion

Main findings

The cluster-robust model explains 5.9% of the variance in log CTR ($R^2 = 0.059$, $F = 214.83$, $p < 0.001$). The low explained variance is expected: click-through rates are driven by many factors unobserved here, including article placement, audience composition, and image quality. The research question concerns the direction and magnitude of headline effects, not prediction.

Figure 1 summarizes the estimated coefficients with 95% cluster-robust confidence intervals. Ten features are statistically significant and robust across all specifications reported in Table 1. Among the negative effects, the presence of an exclamation mark is the strongest single predictor, associated with a 24.7% reduction in CTR ($\hat{\beta} = -0.283$, $p < 0.001$). Collective framing using first-person plural language reduces CTR by 21.6%, and phrasing the headline as a question reduces it by 15.4%. Headlines beginning with *how*, *why*, or *what* and those containing an ellipsis are also associated with lower engagement, at 8.2% and 8.5% respectively. Second-person framing (*you/your*) reduces CTR by 7.0%. Positive sentiment, as measured by the VADER compound score, is negatively associated with CTR ($\hat{\beta} = -0.130$, $p < 0.001$), implying that more emotionally negative or neutral headlines generate more clicks, consistent with a well-documented negativity bias in news consumption [5].

Among positive effects, leading with a digit is associated with a 6.5% increase in CTR, and the presence of any number in the headline with a 5.7% increase. The use of superlative language produces a smaller but significant positive effect of 3.0%. A meaningful control finding is the year trend: holding all headline features constant, CTR declined by approximately 20% per year over the 2013 to 2015 period, reflecting social media changes during that era.

Implications

These results offer actionable guidance for content editors. Exclamation marks, collective framing, and rhetorical questions are commonly associated with “clickbait” style headlines; our estimates suggest that readers had already discounted these signals by 2013 to 2015, responding to them with lower engagement rather than higher. Numeric openings, by contrast, convey a concrete promise of structured information and consistently outperform purely verbal alternatives. The negativity bias finding suggests that editors choosing between emotionally equivalent framings should favor the more urgent or sobering one, provided it is accurate.

Limitations

The estimates are associational rather than causal: unobserved article-level characteristics such as topic salience could confound headline effects, and the sample covers one platform over a narrow window (2013 to 2015), limiting generalizability to other outlets or time periods. CTR measures initial attention only and does not capture reading completion or informational value.

5 Conclusion

This project asked what linguistic and structural characteristics of a headline predict click-through rate in Upworthy’s A/B tests. Across 54,956 headline packages, emotional and rhetorical framing consistently suppresses engagement: exclamation marks, collective first-person language, and question phrasing each reduce CTR by 15 to 25%, and positive sentiment is associated with lower clicks, consistent with a negativity bias in news consumption. Structural numeric cues are the only robust positive signal, with digit-led headlines outperforming verbal alternatives by roughly 6%. These patterns hold across heteroskedasticity-robust, cluster-robust, and impression-weighted specifications and do not vary by content category, suggesting that concrete specificity and negative framing outperform emotional punctuation as engagement signals.

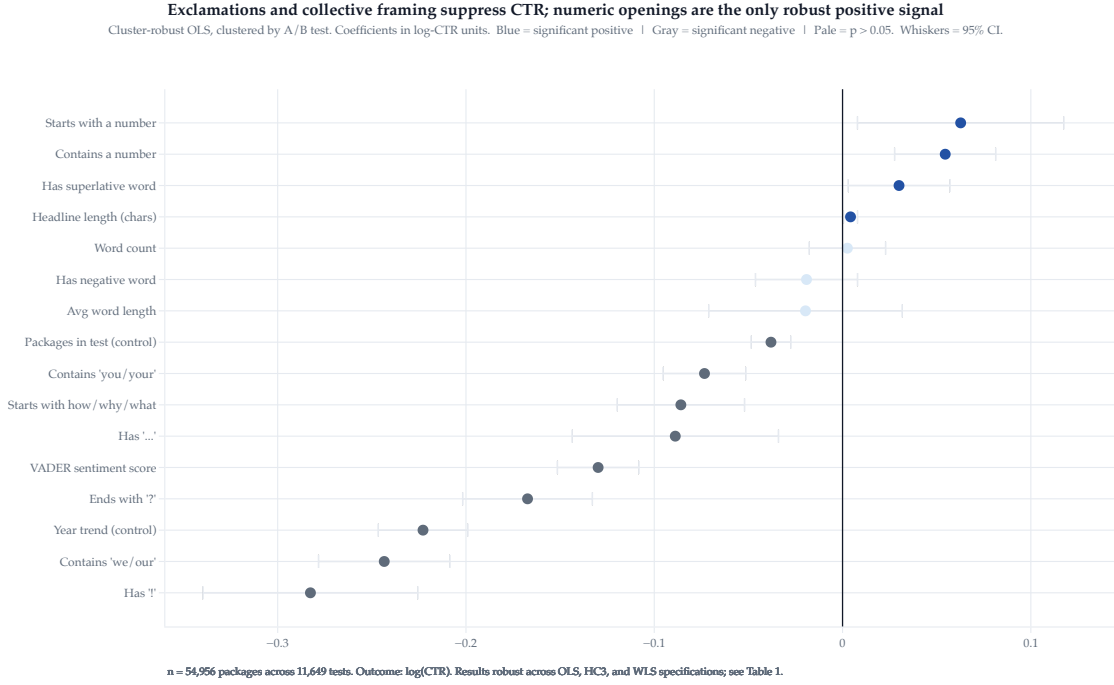


Figure 1: Estimated OLS coefficients with 95% cluster-robust confidence intervals. Blue markers indicate significant positive effects; gray markers indicate significant negative effects; pale markers are not significant ($p > 0.05$). Percentage annotations show implied CTR change for effects larger than 8%. $N = 54,956$ packages, $G = 11,649$ tests.

6 Additional Work

Exploratory analysis and feature engineering

Raw CTR is heavily right-skewed (mean 0.016, median 0.012) with a long upper tail driven by viral packages. The log transformation reduces skewness and stabilizes variance, justifying its use as the outcome. Temporal analysis revealed a decline in mean CTR from 2013 to 2015, motivating the inclusion of a centered year control. Feature correlations were inspected prior to modeling: no pair of headline features exceeded $|r| = 0.35$, ruling out multicollinearity as a concern.

The 14 headline features were engineered from raw text using regular expressions and the VADER lexicon [2]. Binary indicators were preferred over continuous counts for structural features (e.g., presence of “!” rather than count of “!”) because Upworthy headlines rarely repeat punctuation marks, making the binary and count versions nearly identical.

Regression diagnostics

Four standard OLS diagnostics were examined. The residuals-versus-fitted plot showed no systematic non-linear pattern, supporting the linear specification. The normal Q-Q plot tracked the reference line closely in the center with slightly heavier tails, consistent with the presence of a small number of viral outliers; at $N = 54,956$ this does not threaten the validity of normal-approximation inference.

The scale-location plot revealed mild heteroskedasticity, with variance increasing slightly at higher fitted values. Cook’s distance confirmed that no individual package exerts outsized influence on the estimates: fewer than 1% of observations exceeded the conventional threshold of $4/(N - K - 1)$.

These diagnostics motivated the move from OLS to cluster-robust standard errors as the primary inferential tool. HC3 heteroskedasticity-robust standard errors were also computed as an intermediate check; the HC3-to-OLS ratio ranged from 0.95 to 1.03 across all features, indicating that heteroskedasticity alone is negligible and that within-test correlation is the dominant source of non-independence.

Robustness checks and extended models

Weighted least squares. To assess whether results are driven by low-impression packages with noisy CTR estimates, we refit the model weighting each package by its impression count. All ten significant features retained their sign and significance; the largest coefficient shift was 27% for the ellipsis indicator, with all other key features changing by less than 10%.

Content category classification. Headlines were classified into seven content categories (lifestyle, social justice, health, science, politics, environment, economics) using zero-shot inference with a pretrained natural language inference model [6]. A confidence threshold of 0.30 yielded labels for 77.9% of packages. A one-way ANOVA on log CTR across categories produced $F = 254.36$ ($p < 0.001$), confirming that content category is associated with baseline engagement differences. Adding category fixed effects to the main model increased R^2 from 0.059 to 0.092.

Interaction model. We tested whether the sentiment effect varies by content category by adding category-by-VADER interaction terms to the fixed-effects model. The joint F -test for the six interaction terms was $F = 1.26$ ($p = 0.27$), providing no evidence of category-specific moderation. The negativity bias in CTR appears uniform across content types.

A Data Cleaning Details

Starting from 105,551 raw packages in the confirmatory file, we apply three exclusions in sequence. First, packages from single-variant tests are removed since no within-test comparison is possible. Second, we exclude tests in which packages differ across more than one unique image (approximately 35% of tests); CTR differences in those tests confound headline with image choices. Third, we remove test weeks 201326 to 201402, during which a Cloudflare caching misconfiguration broke random assignment, as documented in [1]. These filters together reduce the sample from 105,551 to 54,956 packages.

B Robustness table

Table 1: Robustness of headline coefficient estimates across standard error specifications. Coefficients are identical across columns; only standard errors (in parentheses) and significance change. Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

	OLS	HC3	Cluster	WLS+Cluster
Ends with “?”	−0.167*** (0.012)	−0.167*** (0.012)	−0.167*** (0.018)	−0.158*** (0.019)
Has “!”	−0.283*** (0.023)	−0.283*** (0.023)	−0.283*** (0.029)	−0.283*** (0.030)
Has “...”	−0.089*** (0.021)	−0.089*** (0.020)	−0.089** (0.028)	−0.113*** (0.028)
Starts with a digit	0.063*** (0.018)	0.063*** (0.018)	0.063* (0.028)	0.061* (0.030)
Contains a digit	0.055*** (0.009)	0.055*** (0.008)	0.055*** (0.014)	0.063*** (0.014)
Contains “you/your”	−0.073*** (0.007)	−0.073*** (0.007)	−0.073*** (0.011)	−0.079*** (0.012)
Contains “we/our”	−0.243*** (0.011)	−0.243*** (0.012)	−0.243*** (0.018)	−0.255*** (0.019)
Has superlative word	0.030** (0.009)	0.030** (0.009)	0.030* (0.014)	0.037* (0.014)
Has negative word	−0.019* (0.009)	−0.019* (0.009)	−0.019 (0.014)	−0.017 (0.014)
Starts how/why/what	−0.086*** (0.012)	−0.086*** (0.012)	−0.086*** (0.017)	−0.087*** (0.019)
VADER sentiment score	−0.130*** (0.007)	−0.130*** (0.007)	−0.130*** (0.011)	−0.118*** (0.012)
Headline length (chars)	0.004** (0.001)	0.004** (0.001)	0.004* (0.002)	0.006** (0.002)
Word count	0.003 (0.007)	0.003 (0.007)	0.003 (0.010)	−0.006 (0.011)
Average word length	−0.020 (0.019)	−0.020 (0.019)	−0.020 (0.026)	−0.041 (0.028)
Year trend (control)	−0.223*** (0.006)	−0.223*** (0.006)	−0.223*** (0.012)	−0.193*** (0.012)
Variants per test (control)	−0.038*** (0.002)	−0.038*** (0.002)	−0.038*** (0.005)	−0.043*** (0.006)
N	54,956	54,956	54,956	54,956
R^2	0.059	0.059	0.059	0.052
SE estimator	Homoskedastic	HC3	Cluster	Cluster
Weights	None	None	None	Impressions

References

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